

Mithil Hardik Astik

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EDUCATION

BITS Pilani, Hyderabad Campus

Hyderabad, India

Integrated M.Sc. Physics and B.E. Electrical & Electronics Engineering; CGPA: 6.98

Aug. 2024 – Present

- Relevant study: Quantum Information and Computing, Computational Physics, Quantum Mechanics, Electromagnetism and Optics.

RESEARCH

LHC Olympics 2020 R&D Anomaly-Detection Benchmark

2026

Sole-authored manuscript; simulated collider events

Python, PyTorch, scikit-learn

- Benchmarked 13 unsupervised, semi-supervised and supervised detectors; evaluated contamination, label budget, dijet-mass sculpting and transfer to a held-out three-prong signal topology.
- Deep SAD reached AUC 0.934 ± 0.001 with 100 labelled anomalies and 0.5% training contamination; held-out-topology AUC was 0.824 ± 0.008 . Injected-signal benchmark exceeded 8σ at 1% background efficiency. Simulation study.

Simulation-Based Detection of Optical Injection-Locking Anomalies in TF-

2026

QKD

M. H. Astik and T. S. L. Radhika; manuscript

Python, Lang-Kobayashi dynamics

- Built a stochastic laser and decoy-state twin-field QKD simulator; generated 22 million pulses with randomized FIM, a TWIRL frequency-sweep proxy and 10 held-out synthetic attack mechanisms. Compared semi-supervised Deep SVDD-type and XGBoost detectors.
- At 100 km, pooled zero-day AUROC was 0.925 (Deep SVDD-type) and 0.935 (XGBoost); identified carrier-density modulation as a blind spot (AUROC 0.334/0.299) and non-monotonic transfer across laser settings. No hardware validation.

ML-Based Attack Detection for Twin-Field QKD with Optical Injection Locking

2026

M. H. Astik and T. S. L. Radhika; ICICSCS 2026

Python, machine learning

- Compared classical and quantum classifiers on 40,000 simulated telemetry samples; XGBoost reached 87.52% test accuracy, while drift augmentation reduced simulated false positives from 99.88% to 2.48%.

Reproduction of PIC-Based Distributed-Phase-Reference QKD Simulations

Completed

Paper Reproduction Experiment

Photonic-integrated-circuit Simulations

- Reproduced published simulation curves for PIC-based QKD under detector-blinding, Trojan-horse and backflash attacks.

PROJECTS

APEX-PINN: Physics-Informed Neural PDE Solvers | *Python, PyTorch*

In progress

- Developing coupled fluid, heat-transfer and magnetic-induction PDE solvers; implemented divergence-free parameterizations, adaptive collocation, multi-fidelity training and inverse parameter estimation.

QKD-SecLab | *Python, PyTorch*

In progress

- Developing a modular QKD simulation and detector lab across BB84, MDI-QKD and TF-QKD, with attack/drift models and leakage-aware evaluation.

EXPERIENCE & LEADERSHIP

Quantum Computing Team Lead, IEEE Student Branch, BPHC

Feb. 2025 – Present

BITS Pilani, Hyderabad Campus

India

- Led Qiskit activities; secured an IBM Quantum-sponsored partnership and led Quantumverse Vol. 3 with ATMoS '25.

Summer Intern

May 2026 – Jul. 2026

Beijan, BITS Pilani, Hyderabad

India

- Developed a React hardware-telemetry viewer and configured edge-computing interfaces for GPS-denied navigation.

Freelance Full-Stack Developer

Feb. 2026 – Present

Multiple clients

Remote

- Built a retail inventory and ordering application using Next.js, PostgreSQL and TypeScript.

TECHNICAL SKILLS

Scientific computing: Python, C/C++, NumPy, PyTorch, numerical simulation, PDE modelling, ANSYS Lumerical, FastJet

Machine learning: scikit-learn, XGBoost, LightGBM, SHAP, anomaly detection, MLflow

Quantum: Qiskit, PennyLane, quantum-kernel methods

Software: Git, TypeScript, JavaScript, React, Node.js, FastAPI, PostgreSQL, SQLite